Evaluating the Effectiveness and Barriers of mHealth-Based Interventions Promoting Lifestyle and Behavioral Changes Among Adults with Prediabetes

Course : SAT 5424 Population Health Informatics – Spring 2026

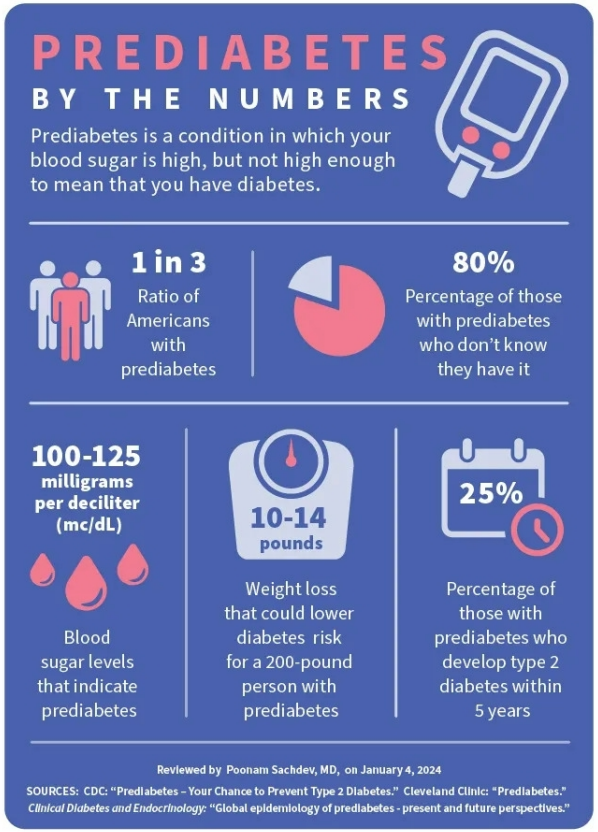
Students: Group 2

- **Sandra Igboabuchukwu**
- **Suha Khalil**

Michigan Technological University

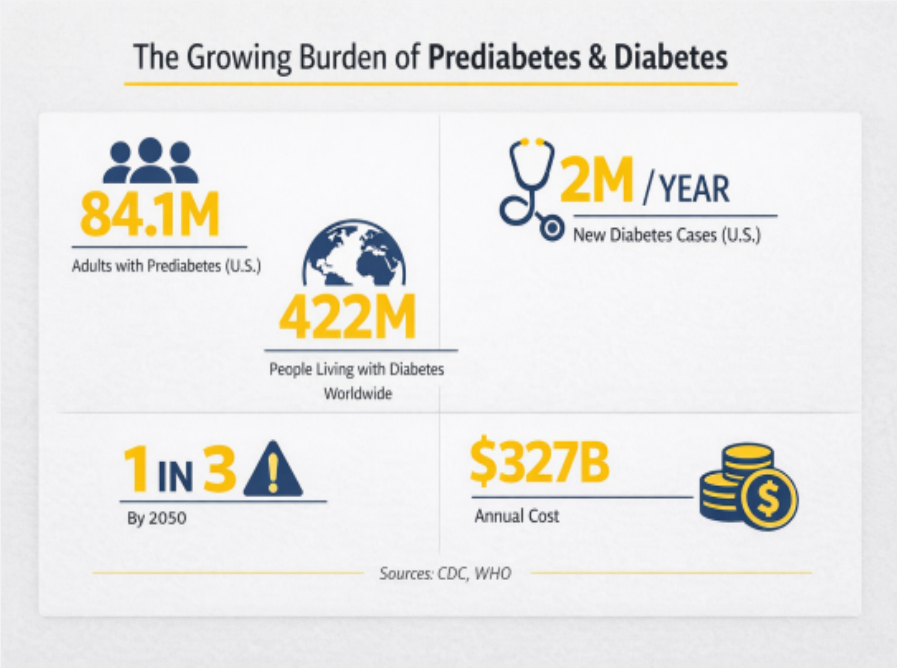
Introduction

1. What is Prediabetes?
2. Why it matters?
3. What is mHealth?
4. What is the connection between mHealth and Prediabetes?



Source: https://www.webmd.com/diabetes/what-is-prediabetes

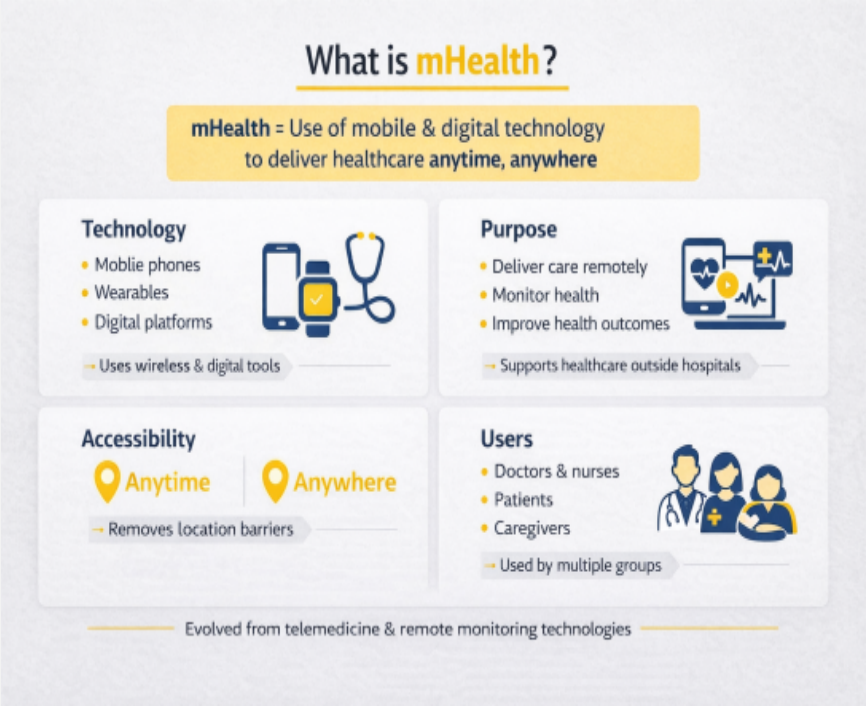
Why Prediabetes matters?



Let's Talk Numbers!

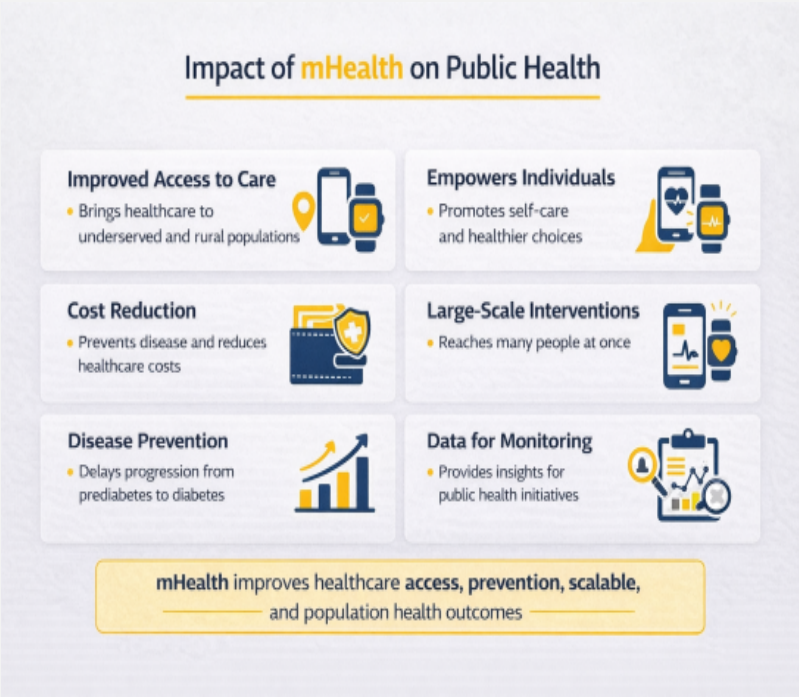
- The U.S. prediabetes population alone is nearly 20% the size of the global diabetes population.
- 80% of people with prediabetes do not know they have it.
- Prediabetes can lead to Diabetes type 2 within 5 years if left untreated
- The $327 billion annual cost in the US reflects not only healthcare expenses but also lost productivity, highlighting the urgent need for cost-effective prevention strategies.

What is m-health



- In-person DPP(Diabets Prevention Program) forms the foundation for dibetes prevention or management.
- Due to limited time health-system barrier like: Limited clinician time and patient barrier like: Low motivation, particpation and adherence the program was not egffective
- Hence, the need for mhealth, that prevents time constraint, scalable and accessible.
- 4 out of 5 Ameriacans have smart-phones
- mHealth is the use of mobile phones to provide public health and medical solutions in a scalable and cost-effective manner.
- Mobile apps, text messaging-based interventions, integrated web programs,for managing lifestyle.

What is the Impact on Public health?



- It improves patient's adherence especially for the DPP(Diabetes Prevention Program)
- Helps with Patient Generated Health Data (PGHD) to prevents other chronic diseases that could be related to Diabetes (e.g Kidney Disease, Cardiovascular problems, Retinopathy)
- It helps to provide access to care especially for health populations in rural regions which is one of the important social determinant of health(SDOH)
- It additionally reduces the health cost burden by allowing lifestyle changes that prevent prediabetic people from becoming type 2 Diabetes patients.

How mHealth Supports Change for a Prediabetic Patient?

1. Daily reminders and motivational messages
2. Physical activity and step tracking
3. Nutrition logging and calorie monitoring
4. Goal Setting and Feedback
5. Remote coaching and telehealth support
6. Social Learning and peer support groups

Barriers and Challenges Related to m-health based intervention for Prediabetes

1. Low long-term engagement / app fatigue
2. Digital literacy differences
3. Privacy and data security concerns
4. Internet/device access inequality
5. Language and cultural barriers
6. Limited personalization in some apps

Related Work on m-health based interventions

1. Chen et al. (2020)

investigated the effects of **mobile-app-based low-carbohydrate dietary guidance** specifically targeting postprandial hyperglycemia in **adults with prediabetes**

Highlights how digital tools can **provide specialized nutritional support to manage blood sugar levels after meals**

2. Sepah et al. (2015)

evaluated a **Web-based diabetes prevention program** using digital therapeutics. Participants achieved significant clinical outcomes.

Average **4.7% reduction in baseline body weight** and a **0.38% reduction in A1c levels** after one year of the intervention.

Related Work on m-health based interventions

3. Muralidharan et al. (2017)

detailed the protocol for a randomized controlled trial of an mHealth technology known as **mDiab***, which was specifically designed for the **prevention and management of type 2 diabetes.**

*12-week program (smartphone app with health coach) support to track weight, diet, and activity

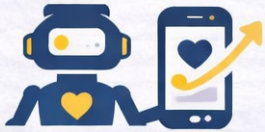
4. Fukuoka et al. (2015)

Explored a **novel diabetes prevention intervention** that combined a mobile application with **pedometer-based tracking.**

Trials specifically for **overweight adults at risk** of developing diabetes, focusing on **physical activity** as a key **behavioral** change

Opportunities for Future Research

AI-Driven Coaching



AI-driven personalized coaching

Gamification

Gamification for retention

Rural Health

Better interventions for rural populations

EHR Integration

Integration with EHR systems

Equity-Focused Design

Equity-focused design for vulnerable groups

Conclusion

- In conclusion, **mHealth** is a fast, accurate, and safe technology for chronic disease prevention. It improves reach, convenience, and scalability. However, we must ensure these solutions are inclusive and fair to avoid widening the 'digital divide'. The path forward requires a convergence of healthcare, data science, and behavioral science to empower individuals and optimize public health spend.

References:

- **1.** Toro-Ramos, T., Michaelides, A., Anton, M., Karim, Z., Kang-Oh, L., Argyrou, C., Loukaidou, E., Charitou, M. M., Sze, W., & Miller, J. D. (2020). Mobile delivery of the diabetes prevention program in people with prediabetes: Randomized controlled trial. *JMIR mHealth and uHealth, 8*(7), e17842. **https://doi.org/10.2196/17842**
- **2.** Arora, Sanjay, et al. "Implementation and Evaluation of an Automated Text Message–Based Diabetes Prevention Program for Adults With Pre-diabetes." *Journal of Diabetes Science and Technology*, vol. 18, no. 5, 2023, **https://doi.org/10.1177/1932296823116**
- **3.** Adu, M. D., Malabu, U. H., Malau-Aduli, A. E. O., Drovandi, A., & Malau-Aduli, B. S. (2020). Efficacy and acceptability of My Care Hub mobile app to support self-management in Australians with type 1 or type 2 diabetes. *Healthcare, 8*(3), 239. **https://doi.org/10.3390/healthcare8030239**

References Cont:

- **4.** Chen, X., et al. (2020). The Effects of Mobile-App-Based Low-Carbohydrate Dietary Guidance on Postprandial Hyperglycemia in Adults with Prediabetes. Diabetes Therapy, 11(10), 2341-2355. https://doi.org/10.1007/s13300-020-00906-x
- **5**. Sepah, S. C., Jiang, L., & Peters, A. L. (2015). Long-term outcomes of a Web-based diabetes prevention program: 2-year results of a single-arm longitudinal study. Journal of Medical Internet Research, 17(4), e92. https://doi.org/10.2196/jmir.4052
- **6**. Muralidharan, S., et al. (2017). Mobile health technology (mDiab) for the prevention of type 2 diabetes: protocol for a randomized controlled trial. JMIR Research Protocols, 6(12), e242. https://doi.org/10.2196/resprot.8644
- **7.** Fukuoka, Y., Gay, C. L., Joiner, K. L., & Vittinghoff, E. (2015). A novel diabetes prevention intervention using a mobile app: a randomized controlled trial with overweight adults at risk. American Journal of Preventive Medicine, 49(2), 223-237. https://doi.org/10.1016/j.amepre.2015.01.003

Thank you!